

Department of Computer and Software Engineering**SE: Machine Learning****Course Instructor:****Date:****Day:****Semester:****Lab 13: Drift Detection in Streaming Data**

Lab Title	CLO	BT	PLO	Weightage
Drift Detection in Streaming Data KS Test, Alerts & Concept Drift				

Name	Roll Number	Report /10	Viva /5	Total /15

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Objectives

- Load and explore a structured transaction dataset with controlled drift
- Simulate streaming data by slicing the dataset into monthly batches
- Detect statistical distribution shifts using threshold-based methods
- Apply the Kolmogorov-Smirnov (KS) test for formal drift detection
- Understand feature drift vs. concept drift
- Observe how concept drift degrades a trained classifier over time

Dataset

The dataset for this lab is generated using the provided Python script `generate_drift_dataset.py`. Running the script produces a file named `transactions_with_drift.xlsx` containing synthetic online payment transactions over a six-month period (January–June 2024).

The file contains three sheets:

Sheet	Description
All_Transactions	3,000 rows across all six months, sorted by timestamp. Use this as your streaming data source.
Baseline_M1_M3	Months 1–3 only (1,500 rows). Use this as the reference distribution for all drift comparisons.
Drift_Summary_Ground_Truth	Per-month true statistics. Consult this only after completing all tasks to verify your results.

The dataset contains the following columns:

Column	Description
transaction_id	Unique transaction identifier
timestamp	Date and time of the transaction
month	Month number (1–6); use this to split data into batches
transaction_amount	Transaction amount in USD; primary drift feature
customer_age	Age of the customer in years
transaction_hour	Hour of day (0–23); shifts toward later hours under drift
device_risk_score	Device risk score (0.0–1.0); increases under drift
is_fraud	Binary fraud label: 1 = fraud, 0 = legitimate
drift_phase	Phase label: <i>Baseline</i> , <i>Feature Drift</i> , or <i>Concept Drift</i>

Problem Statement

Modern machine learning systems operate in continuously changing environments where incoming data may gradually differ from the data used during training.

Examples include:

- Fraud patterns changing over time
- Customer behaviour evolving seasonally
- Sensor calibration drift in IoT systems
- Changing traffic conditions in ride-sharing systems

In this lab, you are a data scientist at an online payment platform. The fraud detection model was trained on transaction data from January to March 2024 (Months 1–3). From April onwards, customer spending behaviour began to shift — transaction amounts increased, customers transacted at later hours, device risk scores rose, and a new wave of fraud emerged with characteristics different from the training period.

Your task is to build a drift detection pipeline that:

- Treats each month as an incoming streaming batch
- Detects distribution shifts in transaction features
- Raises alerts when drift is detected
- Logs drifting batches with supporting statistics
- Uses the KS test for statistical comparison

- Observes concept drift effects on model performance

You will work with a transaction dataset containing:

- Transaction amount
- Customer age
- Transaction hour
- Device risk score
- Fraud label

Part A: Streaming Data Simulation

Task 1: Load Baseline Dataset

Load the `Baseline_M1_M3` sheet and compute the following for each numerical feature (`transaction_amount`, `customer_age`, `transaction_hour`, `device_risk_score`):

- Mean
- Standard deviation
- Minimum and maximum

Also compute the overall fraud rate (proportion of rows where `is_fraud = 1`). Record these values carefully — they serve as your reference statistics for all drift detection tasks.

Task 2: Simulate Streaming Batches

Load the `All_Transactions` sheet and split the data into six monthly batches using the `month` column. Each batch represents one month of incoming transactions. For each batch, compute the mean of `transaction_amount` and the mean of `transaction_hour`. Visualize how these two feature means evolve across the six months using line plots.

Part B: Threshold-Based Drift Detection**Task 3: Mean Shift Detection**

Using the baseline mean and standard deviation of `transaction_amount` computed in Task 1, implement a drift detector that compares each monthly batch mean against the baseline.

Raise a drift alert for a batch if:

$$|\mu_{batch} - \mu_{baseline}| > 2\sigma_{baseline}$$

where:

- μ_{batch} = mean of `transaction_amount` in the current monthly batch
- $\mu_{baseline}$ = mean of `transaction_amount` in Months 1–3
- $\sigma_{baseline}$ = standard deviation of `transaction_amount` in Months 1–3

Apply this check to all six monthly batches and record which months trigger an alert.

Task 4: Drift Logging

For every month that triggers a drift alert in Task 3, store and display the following information:

- Batch ID (month number)
- Drift phase (*Baseline*, *Feature Drift*, or *Concept Drift*)
- Batch mean of `transaction_amount`
- Shift magnitude: $|\mu_{batch} - \mu_{baseline}|$
- Number of standard deviations from baseline: $\frac{|\mu_{batch} - \mu_{baseline}|}{\sigma_{baseline}}$

Display a summary table of all detected drift events. After completing all tasks, verify your results against the `Drift_Summary_Ground_Truth` sheet.

Part C: KS Test

Task 5: Apply KS Test

Use the Kolmogorov-Smirnov (KS) test to compare the `transaction_amount` distribution of the baseline (Months 1–3) against each of the six monthly batches. For each batch, report:

- KS statistic
- p-value

Flag a batch as drifted if the p-value falls below 0.05. Record which months are flagged.

Task 6: Compare Statistical and Threshold Methods

Construct a comparison table with one row per month, showing the mean-shift alert (Task 3) and KS test alert (Task 5) side by side.

Discuss:

- Do both methods agree on which months show drift?
- Which method detects drift earlier?
- Sensitivity, false alarm rate, and robustness of each method

Part D: Concept Drift

Task 7: Train Baseline Model

Train a logistics regression classifier on the `Baseline_M1_M3` data to predict `is_fraud`.

Use the four numerical features: `transaction_amount`, `customer_age`, `transaction_hour`, and `device_risk_score`.

Evaluate on a held-out 20% split of the baseline data and report:

- Accuracy
- F1-score

This model represents the production fraud detector before any drift occurred.

Task 8: Evaluate on Drifted Data

Without retraining, evaluate the same model from Task 7 on each of the six monthly batches separately.

For each batch, record:

- Accuracy
- F1-score

Observe how model performance changes as drift increases across months. Pay particular attention to Month 6, where the fraud rate triples compared to the baseline period.

Part E: Visualization

Task 9: Drift Visualization

Create a single figure with four subplots summarising your findings across all six months:

- Mean of `transaction_amount` per month, with the 2σ alert threshold marked
- KS statistic per month, with the $p = 0.05$ significance boundary marked
- Fraud rate per month (proportion of `is_fraud = 1`)
- Model F1-score per month from Task 8

Shade the background of Months 4–5 to indicate Feature Drift and Month 6 to indicate Concept Drift across all four subplots.

Discussion Questions

- What is the difference between feature drift and concept drift? Use specific months from this dataset to illustrate your answer.
- In Month 4, transaction amounts increased but the fraud rate only rose slightly. Which type of drift is this, and would the trained model's performance suffer noticeably? Why or why not?
- In Month 6, the fraud rate tripled relative to the baseline. Why does this cause the sharpest drop in F1-score even if accuracy does not fall as dramatically?
- Why might a simple mean-shift detector miss a drift event that the KS test successfully catches? Give a concrete example.
- How can a production ML system respond to detected drift automatically? Describe one practical strategy.